From Geometric Inductive Bias to 2D Generative Priors:
A Structured Survey of Deep Learning for 3D Reconstruction from Text and Images

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Abstract

This paper is a structured personal survey compiled during my research collaboration initiation at NAIST Cybernetics and Reality Engineering Lab, under the supervision of Prof. Hideaki Uchiyama. Rather than treating the literature as a neutral sequence of methods, I use it to argue that the recent success of text- and image-conditioned 3D generation did not mainly come from scaling native 3D supervision, but from coupling *geometric inductive bias* with *strong 2D foundation priors*. Classical and early deep 3D methods were constrained by irregular geometry, partial observations, and limited 3D datasets. Neural implicit representations, differentiable rendering, and multi-view supervision mitigated these issues, but the decisive shift, in my reading, occurred when frozen 2D models such as CLIP and diffusion models began supervising 3D optimization through rendered views. I organize the literature along three transitions: from geometry-first supervised reconstruction to implicit neural fields; from view-based 3D reasoning to score-distilled text-to-3D; and from baseline SDS pipelines to reference-guided, retrieval-augmented, and real-time asset generation systems. Across representative methods—including Pixel2Mesh++, NeRF, Zero-1-to-3, DreamFusion, Magic3D, Unique3D, Hi3D, Sculpt3D, ReDream, Phidias, and Meta 3D AssetGen—I argue that modern 3D generation is best understood as *3D optimization under 2D priors*, not as purely native 3D generative modeling. Geometry remains essential, but its role has shifted from being the main source of supervision to being the target of optimization, regularization, and extraction.

1 Introduction

Reconstructing a 3D object from text or image inputs sits at the intersection of computer vision, graphics, and geometric deep learning. I was drawn to the task because it promises intuitive content creation, but the literature also made clear to me why it is structurally difficult: 3D signals are irregular, scans are partial, and common representations such as meshes, point clouds, and voxel grids each trade off fidelity, efficiency, and differentiability [1–5]. Even before text-to-3D emerged, the field already faced a fundamental bottleneck: native 3D datasets are much smaller and noisier than large-scale 2D corpora [6, 7].

As I went through this literature, I came to see this bottleneck as one of the main forces shaping the field's trajectory. Early progress came from injecting *geometric structure*: multi-view consistency, graph-based mesh deformation, implicit fields, and differentiable rendering. Later progress came from injecting *semantic prior*: CLIP-aligned image-text models and diffusion models trained on internet-scale 2D data [8, 9]. The historical outcome, as I understand it, is that successful modern systems rarely learn 3D from scratch. Instead, they optimize a 3D representation so that its rendered views satisfy strong 2D priors.

This perspective also helped me understand why the most effective recent systems do not simply replace geometry with foundation models. They combine: (i) a 3D representation that can be rendered and extracted; (ii) a differentiable bridge from 3D to 2D; and (iii) a large 2D prior that supplies semantics, texture, and realism. In the remainder of this paper, I develop this thesis in a compact, method-centered form.

2 Why Native 3D Learning Is Hard

2.1 Representation mismatch

One lesson I kept encountering is that, unlike images, 3D data does not come with a single canonical grid. Meshes are compact and useful for downstream graphics, but require explicit connectivity. Point clouds are simple and scanner-friendly, but lack surface continuity and topology. Voxels admit regular convolutions, but memory scales cubically with resolution. These difficulties motivated implicit representations, where geometry is modeled as the level set of a learned function

$$\mathcal{S} = \{x \in \mathbb{R}^3 \mid f_\theta(x) = 0\}. \tag{1}$$

DeepSDF and Occupancy Networks showed that continuous neural fields can encode shapes more flexibly than explicit grids, while remaining compatible with extraction procedures such as Marching Cubes [4, 5].

2.2 Data scarcity and supervision mismatch

The second obstacle, in my reading, is supervision. Large 3D datasets exist, but they remain modest compared with

web-scale image-text corpora and often consist of synthetic CAD objects rather than richly captioned, textured, real-world assets [6, 7]. This pushes 3D learning toward either strong priors or proxy supervision. In practice, much of the field evolved by reusing 2D supervision through multi-view renderings, image losses, and later frozen 2D foundation models.

3 Geometry-First Foundations

3.1 Classical reconstruction and implicit surfaces

As I read older reconstruction work, I came to see surface estimation as the conceptual backbone of the field. Hoppe et al. approximated a signed distance field from unoriented points and extracted the zero set as a mesh [1]. Poisson surface reconstruction instead fit an indicator function whose gradient matches an oriented normal field, yielding watertight surfaces that remain a strong classical baseline [3]. These methods still matter to me because they introduced an idea that persists today: represent geometry as a function, then extract a surface from that function.

Neural implicit models extended this paradigm. DeepSDF learned continuous signed distances [4]; Occupancy Networks learned inside/outside probabilities [5]; sign-agnostic and eikonal-based methods improved robustness when normals are noisy or unavailable [10–13]. Across these lines of work, geometry is no longer stored directly but encoded in a differentiable function approximator.

3.2 View-based reasoning and differentiable rendering

A second foundational thread, as I understood it, was to avoid native 3D supervision when possible by projecting the problem into 2D. Multi-view methods render an object from several viewpoints, process each view with 2D CNNs, and fuse features back into a 3D prediction or decision [14–16]. The underlying bridge is differentiable rendering:

$$I_v = \mathcal{R}(\mathcal{S}; \Theta_v), \tag{2}$$

where a 3D scene representation $\mathcal{S}$ and camera parameters Θ_v produce a rendered view I_v. Once gradients can backpropagate through $\mathcal{R}$, image-level losses supervise 3D parameters [17–20].

This idea culminated in NeRF, where a continuous field predicts density and color along rays and images are matched by volumetric rendering [21]. NeRF did not solve text-to-3D by itself, but it delivered a crucial optimization substrate: a differentiable 3D representation expressive enough to absorb supervision from 2D views.

4 The 2D Prior Turn

The decisive conceptual shift, in my view, occurred when 3D optimization began to use 2D foundation models as supervisory or regularizing priors. CLIP-based work such as Text2Mesh already showed that 3D outputs could be optimized so that rendered views align with text in a shared image-text embedding space [8, 22]. But diffusion models changed the scale of what such priors could express.

DreamFusion is the inflection point [9]. It does not train a native 3D generator from paired 3D data. Instead, it optimizes a NeRF so that rendered images receive high probability under a frozen text-conditioned 2D diffusion prior. The central mechanism is Score Distillation Sampling (SDS), whose gradient can be written schematically as

$$\nabla_\theta \mathcal{L}_{\text{SDS}} \approx \mathbb{E}_{t,\epsilon}\left[w(t)\big(\hat{\epsilon}_\phi(x_t, y, t) - \epsilon\big)\frac{\partial x}{\partial \theta}\right], \tag{3}$$

where the diffusion model supplies a denoising direction and the 3D representation parameters θ are updated through the renderer. From this perspective, DreamFusion is less a conventional 3D generative model than a 3D optimization loop driven by a 2D prior.

Magic3D improved fidelity and speed via a coarse-to-fine pipeline: a low-resolution NeRF stage for global geometry, followed by a high-resolution mesh refinement stage for appearance and detail [23]. This two-stage design already reveals a recurring pattern in the literature: 2D priors are powerful but unstable, so systems increasingly rely on structured 3D representations and staged optimization to constrain them.

5 Structured Survey of Representative Methods

Table 1 summarizes representative methods through the lens of the main thesis I developed in this review.

5.1 Geometry-first supervised reconstruction

Pixel2Mesh++ is emblematic of the geometry-first regime. It deforms a mesh template using graph convolutions informed by multi-view image features [16]. I find the method elegant precisely because it shows how much one can recover from structured correspondence and iterative deformation alone, but it also inherits the topological rigidity of mesh deformation. In contrast, DeepSDF and Occupancy Networks are representation-first methods: they gain flexibility by predicting a continuous field, but rely on shape supervision rather than semantic priors [4, 5]. NeRF further improves fidelity and view consistency through volumetric rendering, but still presumes images rather than language as input [21]. I read Zero-1-to-3 as an important bridge between this NeRF-era view-synthesis line and later image-conditioned generation systems such as Unique3D: it uses camera-conditioned diffusion to hallucinate novel views from one image, effectively turning image-conditioned generation into a multi-view supervision problem for downstream 3D lifting [24].

5.2 Text-to-3D by distillation and staged optimization

DreamFusion introduced the dominant recipe for text-to-3D: keep the 2D prior frozen, render the 3D state into

Table 1: Representative methods organized by input modality, 3D representation, supervisory prior, and main contribution.

Method	Input	Representation	Main supervisory prior	Key idea	Typical limitation
Pixel2Mesh++ [16]	Multi-view image	Mesh	Supervised image/mesh pairs	Graph-based mesh deformation from image features	Limited topology flexibility
DeepSDF / OccNet [4, 5]	Shape samples	Implicit SDF / occupancy	Supervised 3D samples	Continuous shape fields instead of fixed grids	Requires shape-specific supervision
NeRF [21]	Multi-view image	Radiance field	Photometric consistency	Differentiable volumetric rendering with view synthesis	Slow optimization; not text-native
Zero-1-to-3 [24]	Single image	Synthesized multi-view image prior	Pose-conditioned 2D diffusion	Zero-shot novel view synthesis that bridges image input and downstream 3D reconstruction	Weak explicit 3D consistency; downstream lifting still needed
Text2Mesh [22]	Text + mesh	Mesh + neural style field	CLIP similarity	Text-driven stylization of a given mesh without 3D pairs	Needs an initial mesh
DreamFusion [9]	Text	NeRF	Frozen 2D diffusion prior (SDS)	Text-to-3D via score distillation, no 3D training pairs	Janus artifacts; slow optimization
Magic3D [23]	Text	NeRF → mesh	2D diffusion prior	Coarse-to-fine pipeline for faster, sharper assets	Multi-stage complexity
Sculpt3D [25]	Text	Sparse 3D prior + radiance field	Diffusion prior + sparse geometric prior	Improves multi-view consistency with explicit sparse guidance	Still optimization-heavy
ReDream [26]	Text	NeRF-like 3D generator	Retrieval-augmented diffusion prior	Uses retrieved 3D priors to stabilize generation	Retrieval quality matters
Unique3D [27]	Single image	Mesh	Multi-view diffusion + reconstruction	Efficient high-quality image-to-mesh pipeline	Strongly conditioned on input view quality
Hi3D [28]	Text / image	3D asset via video-consistent views	Video diffusion prior	Uses sequential/video diffusion for high-res multi-view consistency	Large compute footprint
Phidias [29]	Text / image / 3D refs	3D generator	Reference-augmented diffusion	Controllable generation from multimodal references	Control pipeline is complex
AssetGen [30]	Text	Mesh + texture + PBR materials	Multiview diffusion prior	Jointly targets geometry, texture, and materials for usable assets	High engineering complexity

images, and optimize the 3D state via SDS [9]. Magic3D refined the recipe into a practical multi-stage system that decouples coarse geometry from fine appearance [23]. This family of methods demonstrated that 2D priors can compensate for scarce 3D supervision, but also exposed failure modes such as Janus geometry, oversmoothing, and expensive optimization.

5.3 Beyond vanilla SDS: stronger control signals

Many recent systems, to me, read as attempts to repair the weaknesses of baseline SDS by supplying better-conditioned guidance. Even before retrieval or video diffusion became central, ProlificDreamer showed that one could improve fidelity and diversity by reworking the distillation objective itself rather than abandoning the basic idea of optimizing 3D under a frozen 2D prior [31].

Reference-guided generation. Phidias conditions on text, images, and even 3D references, showing that controllability improves when guidance is not purely textual [29].

Retrieval-aware guidance. ReDream already suggested to me that explicit geometric memory could stabilize distillation; I return to this idea in the next section [26].

Sparse or explicit geometric priors. Sculpt3D injects sparse 3D structure to improve consistency across views [25]. Here again, better 3D outputs arise not by abandoning geometry, but by constraining the optimization landscape.

Video and multiview diffusion. Hi3D and Unique3D leverage video-consistent or multi-view diffusion to produce better cross-view agreement [27, 28]. The implicit message, as I see it, is that the real difficulty is not generating one plausible image of an object, but generating a family of images that correspond to the same 3D entity.

Production-ready assets. Meta 3D AssetGen targets not only geometry, but also texture and physically based materials, bringing the field closer to usable graphics assets [30]. This matters because evaluation based purely on rendered appearance often hides the gap between a pretty render and a deployable 3D asset.

6 Retrieval-Augmented 3D Generation

As I compared ReDream, Sculpt3D, and Phidias, I became convinced that "retrieval-augmented 3D generation" actually names two different interventions. In retrieval-augmented *optimization*, retrieved assets regularize score distillation itself: ReDream uses retrieved 3D structure

to bias the variational objective, while Sculpt3D uses retrieved references to anchor sparse rays and keypoints during optimization [25, 26]. In retrieval-augmented *generation*, by contrast, a feedforward model is conditioned on retrieved or user-provided references at inference or training time; Phidias is the clearest example in this survey, because the retrieved 3D reference is part of the generative conditioning path rather than an external regularizer wrapped around SDS [29].

This distinction matters to my broader thesis because retrieval behaves like a non-parametric approximation of a native 3D prior. Instead of learning all geometric regularity into model weights through large-scale 3D pretraining, the system queries an external memory of shapes and imports local structure on demand. I therefore see retrieval as a middle ground between pure 2D distillation, where the optimizer must invent geometry under weak 3D constraints, and full native 3D pretraining, which would internalize such structure parametrically.

My notes also made the optimization story more concrete. Retrieval helps when it narrows the solution space without collapsing prompt-specific variation, but it fails when the retrieved object is semantically close yet structurally misaligned. In Phidias, the small retrieval database and semantic-similarity lookup can return references with the wrong pose or part layout, which is exactly why mechanisms such as dynamic reference routing and adjustable conditioning strength are necessary. In Sculpt3D, sparse geometric supervision is a design choice that I read as deliberately softer than dense imitation: if the retrieved prior is too dominant, optimization locks onto the wrong topology, especially for prompts that exceed the support of the 3D dataset. ReDream points to the complementary failure mode: even good retrieval improves SDS only at notable computational cost, because the system must still render many views and reconcile retrieved geometry with the 2D diffusion prior. What retrieval changes, then, is not the need for optimization, but the conditioning of that optimization.

7 Cross-Cutting Design Axes

7.1 Representation is still destiny

Even in the era of foundation models, 3D representation strongly shapes performance. Meshes are editable and deployable but topologically rigid. Radiance fields are expressive but expensive to optimize and extract. Gaussian splatting improves rendering efficiency and real-time behavior, but raises new questions about watertight geometry and downstream editing [32]. Gaussian splatting concretely shifts the efficiency-quality tradeoff because it replaces expensive volumetric integration with rasterization of anisotropic primitives, so high-fidelity view feedback becomes much cheaper and closer to interactive rates [32]. In my reading, this makes splats especially attractive as optimization carriers or refinement stages, since one can iterate on appearance far faster than with vanilla radiance fields. But splats do not define a watertight surface in the way an SDF or occupancy field does, so reliable mesh extraction still usually requires an auxiliary surface model or post hoc regularization; that is why deployable geometry remains an open issue for splat-based assets. The recent literature therefore converges on hybrids: optimize in a flexible field, then distill or extract to a mesh or splat-based asset.

7.2 The 2D prior supplies semantics; geometry supplies consistency

My main thesis can now be stated more precisely. The strongest 2D models contribute semantic richness, appearance priors, and text alignment. Geometry contributes view consistency, object permanence, extraction, and editability. Neither component alone is sufficient. Pure geometry-first systems underperform semantically because 3D supervision is limited. Pure 2D priors collapse without geometric constraints because they do not natively enforce a single coherent 3D explanation. The success of recent systems comes from optimizing a geometric state under semantic pressure from 2D priors.

7.3 Optimization remains the bottleneck

A striking fraction of modern text-to-3D work is still optimization-based rather than feed-forward. To me, this reveals both the strength and the weakness of current methods. The strength is modularity: one can plug a powerful frozen prior into a new 3D representation. The weakness is that optimization is slow, brittle, and prone to local minima. Even aggressive acceleration strategies such as Instant-NGP's multiresolution hash encoding reduce the raw cost of radiance-field optimization without removing the more basic problem that each prompt still needs a difficult per-instance search through an underconstrained 3D landscape [33]. Methods that add retrieval, multiview consistency, reference conditioning, or staged refinement are all, in effect, attempts to improve the conditioning of this optimization problem.

8 Open Problems

Four issues remain central.

View consistency and geometry correctness. Janus artifacts and inconsistent backside geometry are still not fully solved, especially for text-only generation.

Evaluation. Metrics such as CLIP similarity reward prompt alignment but say little about watertightness, editability, topology, or physical usability. Render quality is not enough.

Native 3D priors. The field still depends heavily on 2D supervision. A major open question is whether large-scale 3D pretraining can eventually reduce reliance on score distillation from 2D models.

Controllable, editable assets. Real applications demand decomposition into geometry, material, lighting, semantics, and parts. Systems that generate only a plausible view-consistent shell remain insufficient for graphics

pipelines.

9 Conclusion

The recent history of 3D reconstruction from text and images is best understood as a redistribution of roles between geometry and semantics. Early work relied on geometric inductive bias because native 3D learning was constrained by irregular structure and scarce data. Neural implicit representations and differentiable rendering then created a bridge through which 2D supervision could shape 3D outputs. Diffusion-era systems completed the turn: strong 2D priors became the main source of semantic supervision, while geometry became the space in which those priors are made mutually consistent.

This perspective explains why modern progress has been fastest not when researchers abandon geometry, but when they use it strategically. The most successful systems are neither purely 3D nor merely lifted 2D generators. They are hybrid optimization systems in which 3D representations, renderers, and extraction procedures are disciplined by large 2D priors. For a compact student review, I believe that is the central lesson of the field.

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